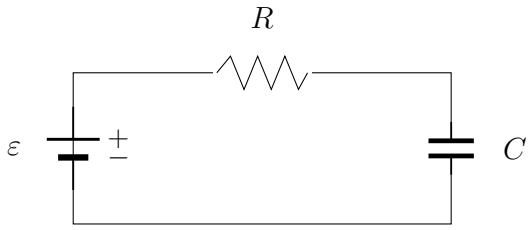
Circuits Review: Capacitors & Kirchhoff's Rules (completed)

Goal.

- See that the loop rule applies to a capacitor exactly like it does to a resistor or a battery — it just contributes a Q/C term.
- Use that at *some instant during charging* (Problems 1–2), where the general relation $\varepsilon = IR + Q/C$ holds at every moment, not just $t = 0$ or $t = \infty$.
- Use it again at *steady state* (Problems 3–4), where a fully charged capacitor carries no current — and see that this sometimes makes a series resistor's value completely irrelevant, and sometimes doesn't.

1. Problem 1 — charge at an instant

A capacitor C is charging through a resistor R, connected to a battery ε, all in a single loop. At a certain moment, while charge is still building up, the current through the resistor is I.



Write the loop rule for this circuit (valid at *any* instant, charging or not) and solve for the charge on the capacitor at this moment.

The loop rule, one resistor and one capacitor:

$$\varepsilon - IR - \frac{Q}{C} = 0.$$

$$\boxed{Q = C(\varepsilon - IR)}$$

Now with $\varepsilon = 9.0$ V, $R = 2.0$ kΩ, $C = 4.0$ μF, $I = 1.5$ mA:

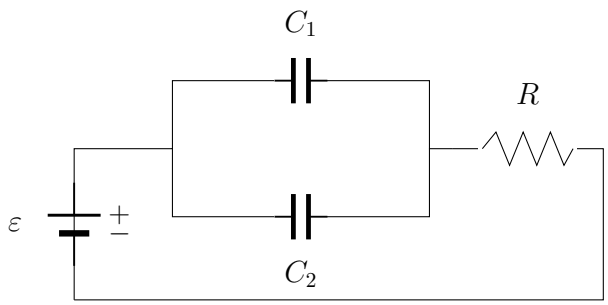
$$Q = (4.0\ \mu\text{F})\big[9.0\ \text{V} - (1.5\ \text{mA})(2.0\ \text{k}\Omega)\big] = (4.0\ \mu\text{F})(6.0\ \text{V}) = 24\ \mu\text{C}$$

Check. This current, $I = 1.5$ mA, is less than the circuit's very first current, $I_0 = \varepsilon/R$. What is I_0 here, and why must the given I always be smaller than it?

$I_0 = \varepsilon/R = 9.0\ \text{V}/2.0\ \text{k}\Omega = 4.5$ mA. The charging current only ever *decays* from I_0 (as the capacitor fills and starts opposing it) — it never starts small and grows. So any instant after $t = 0$ must have $I < I_0$, as it does here.

2. Problem 2 — same idea, with two capacitors in parallel

Two capacitors, $C_1 = 4.0\ \mu\text{F}$ and $C_2 = 6.0\ \mu\text{F}$, are wired in parallel with each other. That parallel pair is in series with a resistor $R = 2.0\ \text{k}\Omega$ and a battery $\varepsilon = 15\ \text{V}$.



At a certain moment during charging, the charge on C_1 is measured to be $Q_1 = 20\ \mu\text{C}$. Find (a) the voltage across the parallel pair at that moment, (b) the current through R, and (c) the charge Q_2 on C_2.

Parallel elements share the same voltage, so start from C_1 alone:

$$\Delta V = \frac{Q_1}{C_1}.$$

Then the loop rule gives the current, and Q_2 follows the same way Q_1 did:

$$\varepsilon - IR - \Delta V = 0 \qquad Q_2 = C_2\,\Delta V.$$

(a)

$$\Delta V = \frac{20\ \mu\text{C}}{4.0\ \mu\text{F}} = 5.0\ \text{V}$$

(b)

$$I = \frac{\varepsilon - \Delta V}{R} = \frac{15\ \text{V} - 5.0\ \text{V}}{2.0\ \text{k}\Omega} = 5.0\ \text{mA}$$

(c)

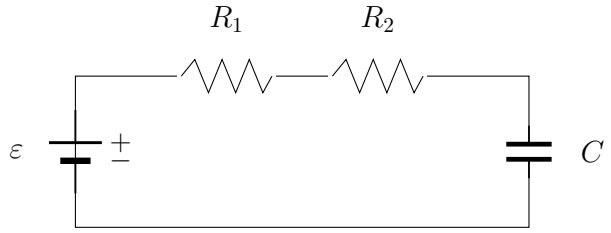
$$Q_2 = (6.0\ \mu\text{F})(5.0\ \text{V}) = 30\ \mu\text{C}$$

Check. C_1 and C_2 have different capacitances but the *same* charge-to-voltage behavior was used for both in part (c). What justifies using one formula, $Q = C\,\Delta V$, for each capacitor separately here, instead of needing some combined parallel-capacitance step?

They're in parallel, so by definition they share the same ΔV — there's nothing to combine first. (If the question had asked for the charge drawn from the *battery*, that total would need $C_\text{p} = C_1 + C_2$; but each capacitor's *own* charge only ever needs its own C and the shared ΔV.)

3. Problem 3 — steady state, resistor values don't matter

A battery $\varepsilon = 10$ V is in series with two resistors, $R_1 = 1.5$ kΩ and $R_2 = 2.5$ kΩ, and a capacitor $C = 3.0$ μF — all in one single loop. The circuit has been connected a long time (steady state).



What is the charge Q on the capacitor?

At steady state the fully charged capacitor carries no current — and since this is a single loop, *nothing else* can carry current either: $I = 0$ everywhere. Both resistors drop $IR = 0$, no matter what R_1 and R_2 actually are.

$$\varepsilon - (0)R_1 - (0)R_2 - \frac{Q}{C} = 0 \quad \Longrightarrow \quad \varepsilon = \frac{Q}{C}.$$

$$\boxed{Q = C\varepsilon}$$

Now with the numbers above:

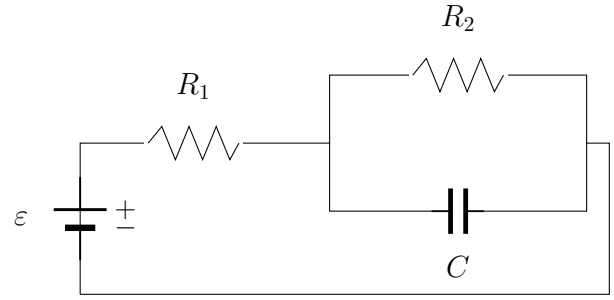
$$Q = (3.0\ \mu\text{F})(10\ \text{V}) = 30\ \mu\text{C}$$

Check. Suppose R_1 and R_2 had been given totally different values. Would Q change?

No — not at all. R_1 and R_2 never appear in the boxed result; they dropped out the moment $I = 0$. Any series resistor (or chain of them) in a single loop like this one becomes completely irrelevant to the final charge once the capacitor is fully charged. This is the opposite situation from Problem 4, where a resistor's value very much does matter — the difference is whether current has somewhere else to go.

4. Problem 4 — steady state, where resistor values do matter

A battery $\varepsilon = 12$ V is in series with a resistor $R_1 = 2.0$ kΩ. That combination feeds a parallel pair: a second resistor $R_2 = 4.0$ kΩ and a capacitor $C = 5.0$ μF. The circuit has reached steady state.



Find (a) the current through R_1 (and through R_2), and (b) the charge on the capacitor.

This time the capacitor being open at steady state does *not* kill all the current — R_2 gives it another way around. Current flows through the loop ε–R_1–R_2 instead:

$$\varepsilon - IR_1 - IR_2 = 0 \quad \Longrightarrow \quad I = \frac{\varepsilon}{R_1 + R_2}.$$

The capacitor is in parallel with R_2, so it ends up at R_2's voltage:

$$\Delta V_C = IR_2 \qquad Q = C\,\Delta V_C.$$

(a)

$$I = \frac{12\text{ V}}{2.0\text{ k}\Omega + 4.0\text{ k}\Omega} = 2.0\text{ mA}$$

(b)

$$\Delta V_C = (2.0\text{ mA})(4.0\text{ k}\Omega) = 8.0\text{ V} \qquad Q = (5.0\ \mu\text{F})(8.0\text{ V}) = 40\ \mu\text{C}$$

Check. In Problem 3, every resistor's value was irrelevant to Q. Here, R_1 and R_2 both clearly matter. What is the one structural difference between the two circuits that explains this?

In Problem 3 the capacitor is the *only* path back to the battery, so blocking it blocks the whole loop, $I = 0$ everywhere, and no resistor drop survives to matter. Here R_2 sits *in parallel* with the capacitor, giving current a second route that doesn't go through C — so current keeps flowing through R_1 and R_2 even once the capacitor is fully charged, and their values set exactly how much voltage lands across it.